Conceptual
Design Review

USC Facilities:

Rainwater Barrel Project

Project Background

- **Green Quad:** First LEED-certified building on campus, focuses on sustainability.
- **USC Office of Sustainability:** Promotes eco-friendly initiatives and resource conservation.

Project Background

Industry Relevance:

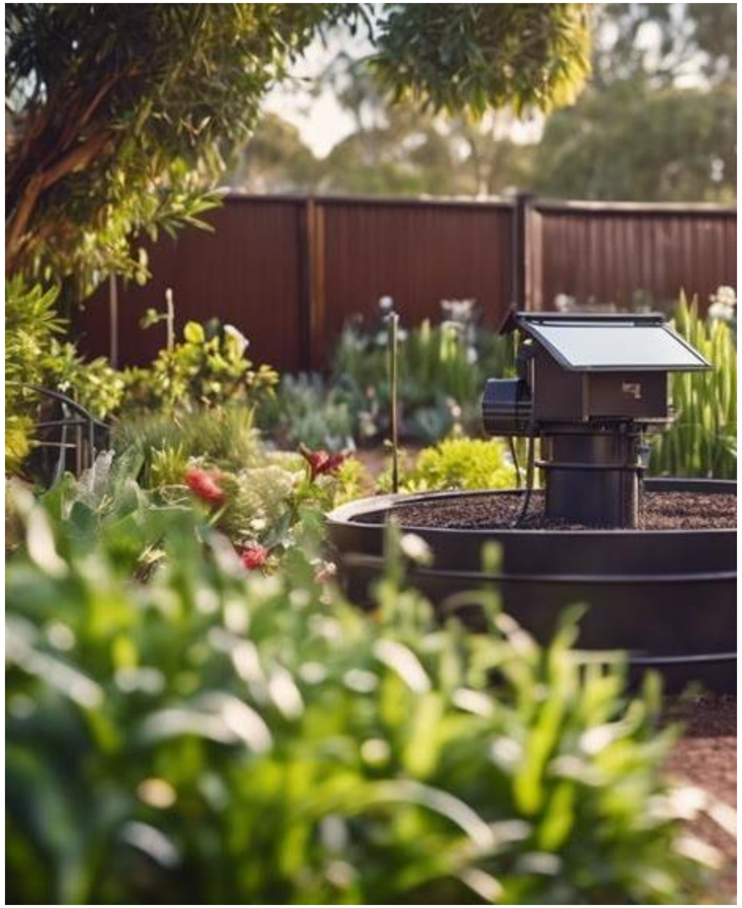
- Aligns with sustainable urban agriculture practices.

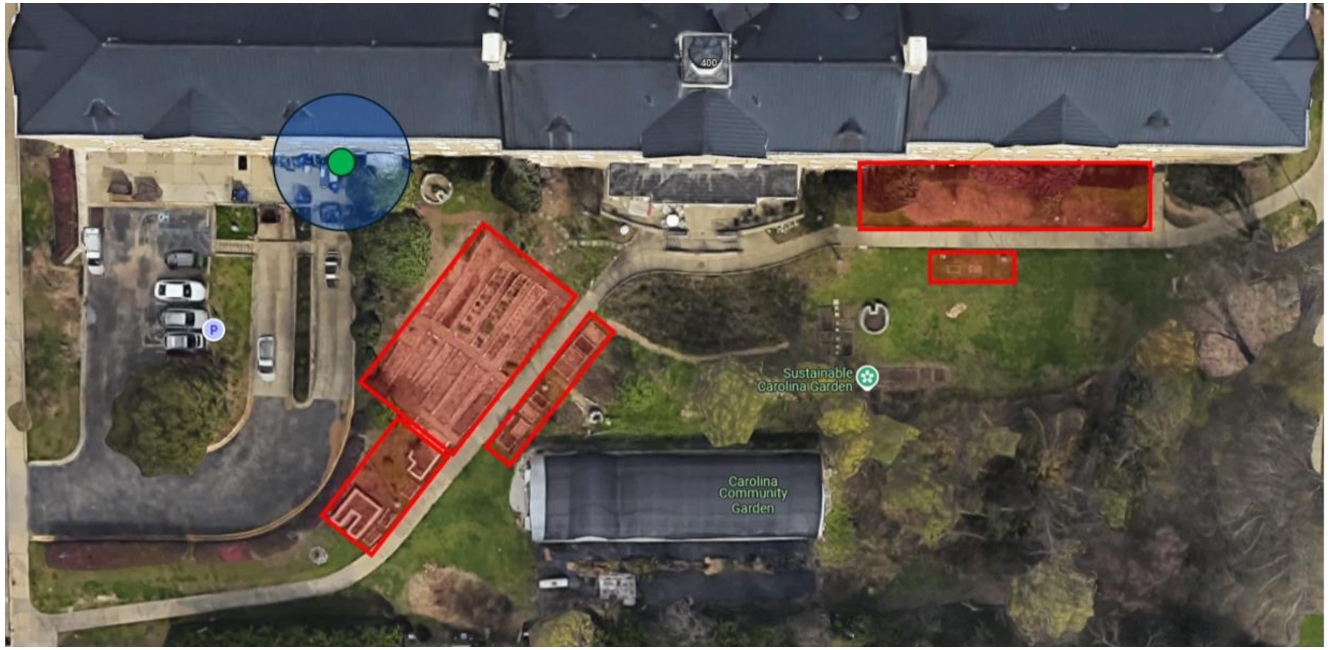
Connection to Challenges:

- Addressing efficient water delivery while maintaining environmental stewardship.

Team Members:

- Tanner Harmon, William Wenzel, Presston McConnell, Jacob Vaught





Current Watering Solution

Water is transferred manually using small watering cans, taking upwards of 1-minute, which is inconvenient for daily garden needs.

Customer Needs [H1]

Current system:

- Manual watering with 2-gallon cans (55 seconds per fill).
- Labor-intensive and inefficient for frequent use.

Stakeholder observations:

- Need for faster, easier water transfer.
- Desire for an eco-friendly solution that aligns with sustainability goals.
- Preference for a low-maintenance system usable by anyone.

User goals:

- Consistent water delivery with minimal physical effort.
- Flexibility for varying schedules and garden needs.

Table 2: CTQ needs prioritization.

Need	#1	#2	#3	#4	Sum
1.1. The system should minimize the time required to transfer water from the tank to the garden.	3	3	4	3	13
1.3. Water distribution should cover the entire garden area efficiently.	2	2	3	2	9
1.6. Water output should be consistent and reliable during the watering process.	2	2	1	1	6
1.12. The system should be easy to activate and deactivate when needed.	0	1	1	1	3
1.15. The system should minimize the physical effort required to operate it.	1	1	0	1	3
2.1. The system should provide sufficient pressure for effective water delivery.	2	2	4	2	10
2.10. The system should provide adequate flow rate for all watering methods.	1	2	3	2	8
5.10. The system should be designed to minimize noise pollution.	1	1	4	1	7
7.1. The system should comply with all safety standards.	2	1	3	1	7

Table 3: Ranking of Critical to Quality (CTQ) needs.

Need	#1	#2	#3	#4	Sum
1.1. The system should minimize the time required to transfer water from the tank to the garden.	6	6	6	6	24
1.3. Water distribution should cover the entire garden area efficiently.	3	4	2	5	14
2.1. The system should provide sufficient pressure for effective water delivery.	4	5	5	3	17
2.10. The system should provide adequate flow rate for all intended watering methods.	5	3	4	4	16
5.10. The system should be designed to minimize noise pollution.	2	2	3	2	9
7.1. The system should comply with all relevant safety standards.	1	1	1	1	4

Critical to Quality Needs

1. Efficiency
2. Pressure
3. Noise
4. Sustainability
5. Safety

Product Mission Statement

Mission Statement:

Create a water distribution system that transfers water quickly, provides sufficient pressure for effective delivery, ensures adequate flow for all watering methods, covers the entire garden area efficiently, and complies with relevant safety standards.

Key Objectives:

Reduce water transfer time.

Deliver consistent water pressure across all garden zones.

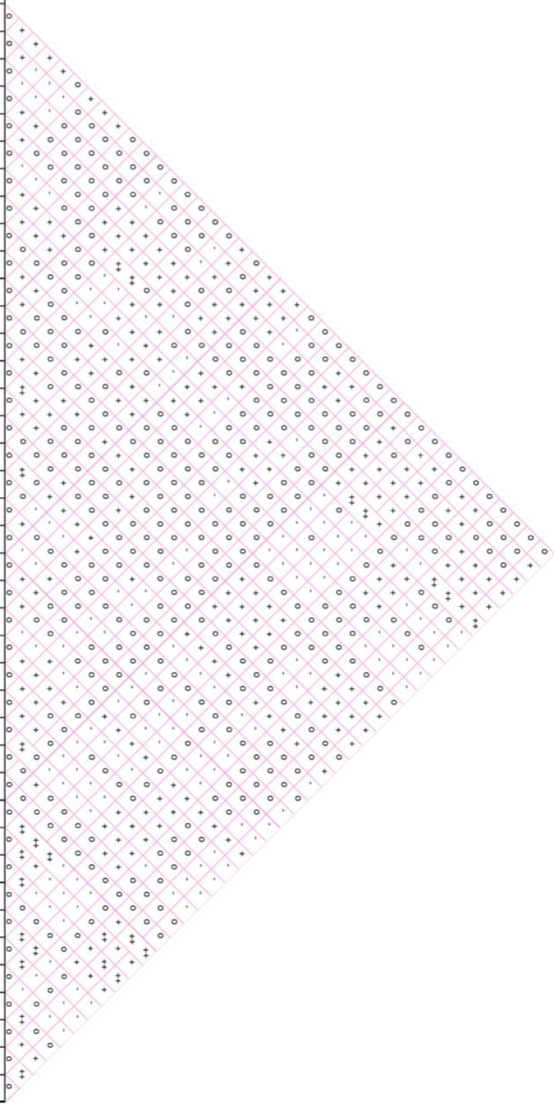
Ensure simplicity and low maintenance for long-term usability.

Incorporate sustainable and recyclable materials.

Specifications: Needs vs Metrics[H2]

- **Top Prioritized Needs:**
  - Quick and efficient water transfer
  - Sufficient water pressure
  - Minimize manual labor through user-friendly operation.
  - Minimize environmental impact with eco-friendly materials.
  - Durability to last at least 10 years under outdoor conditions.
- **Metrics Tracked:**
  - Water transfer rate, flow rate consistency, operational effort time, noise level

Rank	%	Raw	CR21	CR20	CR19	CR18	CR17	CR16	CR15	CR14	CR13	CR12	CR11	CR10	CR9	CR8	CR7	CR6	CR5	CR4	CR3	CR2	CR1	CR#/TR#	Units
Importance			4	3	4	3	4	4	3	3	4	3	4	4	3	3	4	3	4	4	4	5	5		
10	3.22%	136	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	1	3	3	1	3	9	Water Transfer Rate	L/min
10	3.22%	136	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	1	3	3	1	3	9	Time to Water Garden	Minutes
20	2.32%	98	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1	3	3	0	9	3	Distribution Uniformity	%
20	2.32%	98	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1	3	3	0	9	3	Soil Moisture Variation	% water
16	2.86%	121	1	1	1	1	1	1	1	1	1	1	1	1	1	3	3	0	1	1	9	1	1	Manual Operations Count	Count
16	2.86%	121	1	1	1	1	1	1	1	1	1	1	1	1	1	3	3	0	1	1	9	1	1	Operational Effort Time	Minutes
27	1.99%	84	1	1	1	1	1	1	1	1	0	1	1	1	0	1	3	0	1	1	3	1	1	Physical Exertion Level	Borg Scale
5	3.93%	166	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	3	3	9	1	3	9	Water Pressure at Delivery	kPa
2	4.50%	190	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	3	9	9	1	9	3	Flow Rate per Method	L/min
23	2.15%	91	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	9	3	1	3	3	Flow Rate Consistency	% variation
37	1.16%	49	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	1	1	1	1	1	Performance at Varying Levels	L/min and kPa
38	0.92%	39	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	3	0	0	Minimum Operational Water Level	Liters
31	1.56%	66	0	0	0	0	0	0	0	0	0	0	0	0	3	3	9	0	0	0	3	0	0	Learning Time for New Users	Minutes
22	2.30%	97	1	1	1	1	1	1	1	1	1	1	1	0	3	3	9	0	0	0	1	0	0	User Error Rate	Errors/op cycle
3	4.07%	172	1	1	1	1	1	1	1	1	1	1	1	1	3	9	9	1	1	1	9	1	1	Control Interface Count	Count
3	4.07%	172	1	1	1	1	1	1	1	1	1	1	1	1	3	9	9	1	1	1	9	1	1	Operation Initiation Time	Seconds
26	2.01%	85	1	1	1	1	1	1	1	1	1	1	1	1	9	1	3	0	0	0	0	0	0	Instruction Steps Count	Count
19	2.48%	105	1	1	1	1	1	1	1	3	1	3	3	1	9	1	3	0	0	0	0	0	0	Instruction Clarity Rating	Likert scale
23	2.15%	91	1	1	1	1	1	1	1	3	0	3	3	9	0	0	0	0	0	0	0	0	0	Maintenance Frequency	Times per year
6	3.69%	156	3	3	3	3	3	3	0	3	0	3	9	9	0	0	0	0	0	0	0	0	0	Maintenance Duration	Minutes
14	3.12%	132	3	3	3	3	3	3	0	3	0	3	9	3	0	0	0	0	0	0	0	0	0	Expected Lifespan	Years
18	2.84%	120	3	3	3	3	3	3	0	1	0	9	3	3	0	0	0	0	0	0	0	0	0	Warranty Period	Years
6	3.69%	156	3	3	3	3	3	3	0	1	9	9	3	3	0	0	0	0	0	0	0	0	0	Troubleshooting Time	Minutes
29	1.96%	83	1	1	1	1	1	1	1	1	9	1	1	3	0	0	0	0	0	0	0	0	0	Diagnostic Tools Availability	Yes/No
25	2.04%	86	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	1	1	1	1	1	1	Eco-Friendly Materials %	%
30	1.85%	78	0	0	0	0	0	0	0	9	3	0	0	1	1	1	1	1	1	1	1	1	1	Potable Water Savings	Liters/month
33	1.28%	54	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	9	0	0	0	0	0	Rainwater Utilization Rate	%
33	1.28%	54	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	9	0	0	0	0	0	Compatibility Status	Yes/No
40	0.64%	27	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Operating Noise Level	dB
8	3.27%	138	1	1	1	1	1	9	1	1	1	1	1	1	3	3	3	1	1	1	3	1	1	Initial Implementation Cost	$
8	3.27%	138	1	1	1	1	1	9	1	1	1	1	1	1	3	3	3	1	1	1	3	1	1	Cost per Garden Area	$/m²
35	1.23%	52	0	0	0	0	9	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	Annual Operational Cost	$/year
35	1.23%	52	0	0	0	0	9	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	Total Cost of Ownership	$
32	1.30%	55	0	0	0	9	3	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	Cost per Year of Lifespan	$/year
1	5.04%	213	0	0	0	9	1	1	1	1	1	1	1	1	1	1	1	9	9	3	3	9	3	Water Use Efficiency	%
12	3.17%	134	1	9	9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Standards Compliance Count	Count
12	3.17%	134	1	9	9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Certification Status	Yes/No
15	2.93%	124	9	3	3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Hazard Count	Count
27	1.99%	84	9	3	3	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	Risk Assessment Score	low/hi
38	0.92%	39	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	Operational Temperature Range	°C



House Of Quality

Target vs Fallback Specifications

Target Specifications:

Flow rate: 50–60 L/min.

Water pressure: 300–400 kPa.

Noise: ≤50 dB at 1m.

Operating cost: ≤$50/year.

Expected lifespan: 10+ years.

Fallback Specifications:

Flow rate: 30–49 L/min.

Water pressure: 200–300 kPa.

Noise: ≤70 dB at 1m.

Operating cost: ≤$100/year.

Expected lifespan: 5 years.

Trade-offs Identified:

Higher flow and pressure may increase energy consumption.

Noise reduction could raise costs.

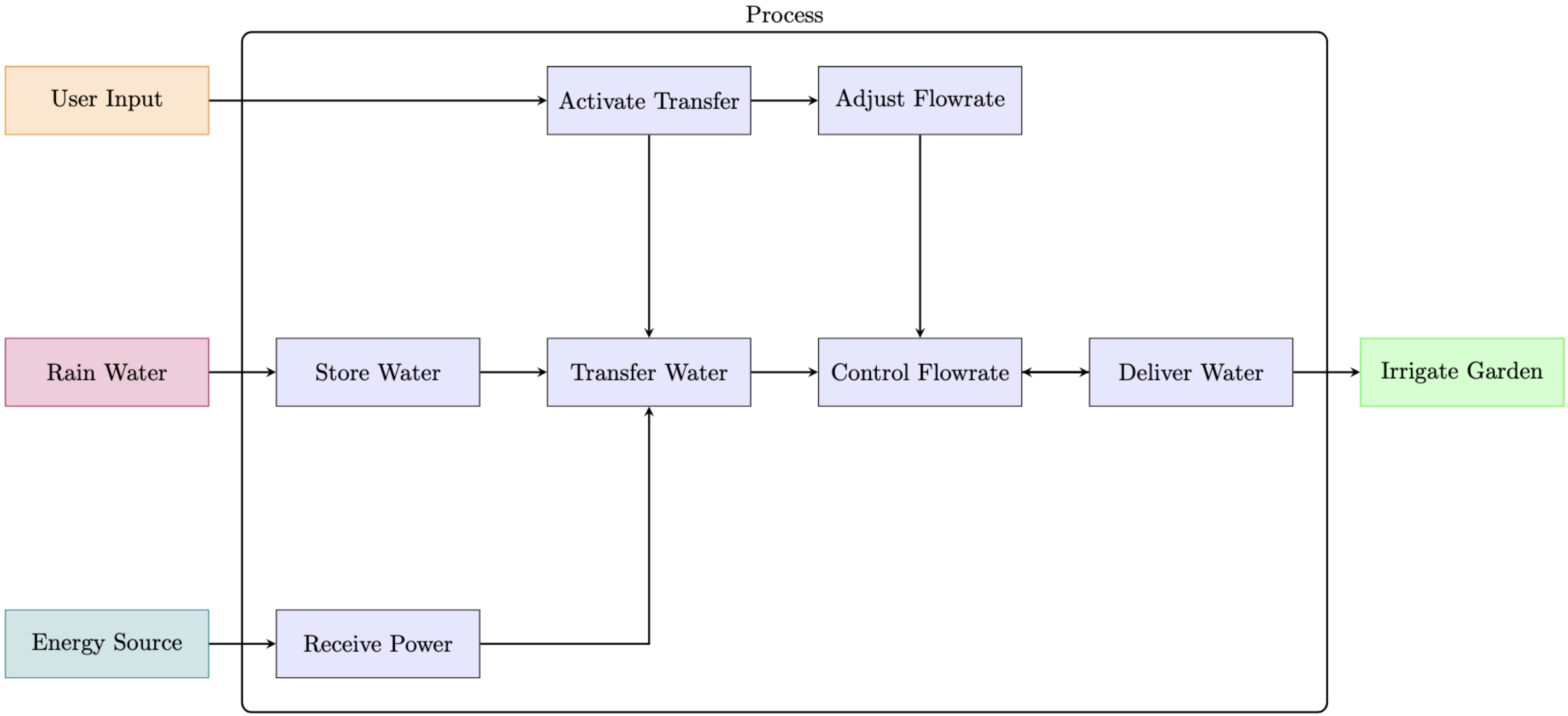
Functional Decomposition[H3]

- Functional decomposition breaks a complex system into smaller, manageable functions to simplify analysis and design.
  - Identifies all system operations needed to achieve the project goals.
  - Helps pinpoint critical components and dependencies.
  - Ensures no functionality is overlooked during the design process.
  - Highlights interactions between subsystems.

Essential Functions

- Receiving Power
- Activation of Transfer
- Adjusting Flow Rate
- Controlling Flow Rate
- Transferring Water
- Storing Water
- Delivering Water
- Irrigating the Garden

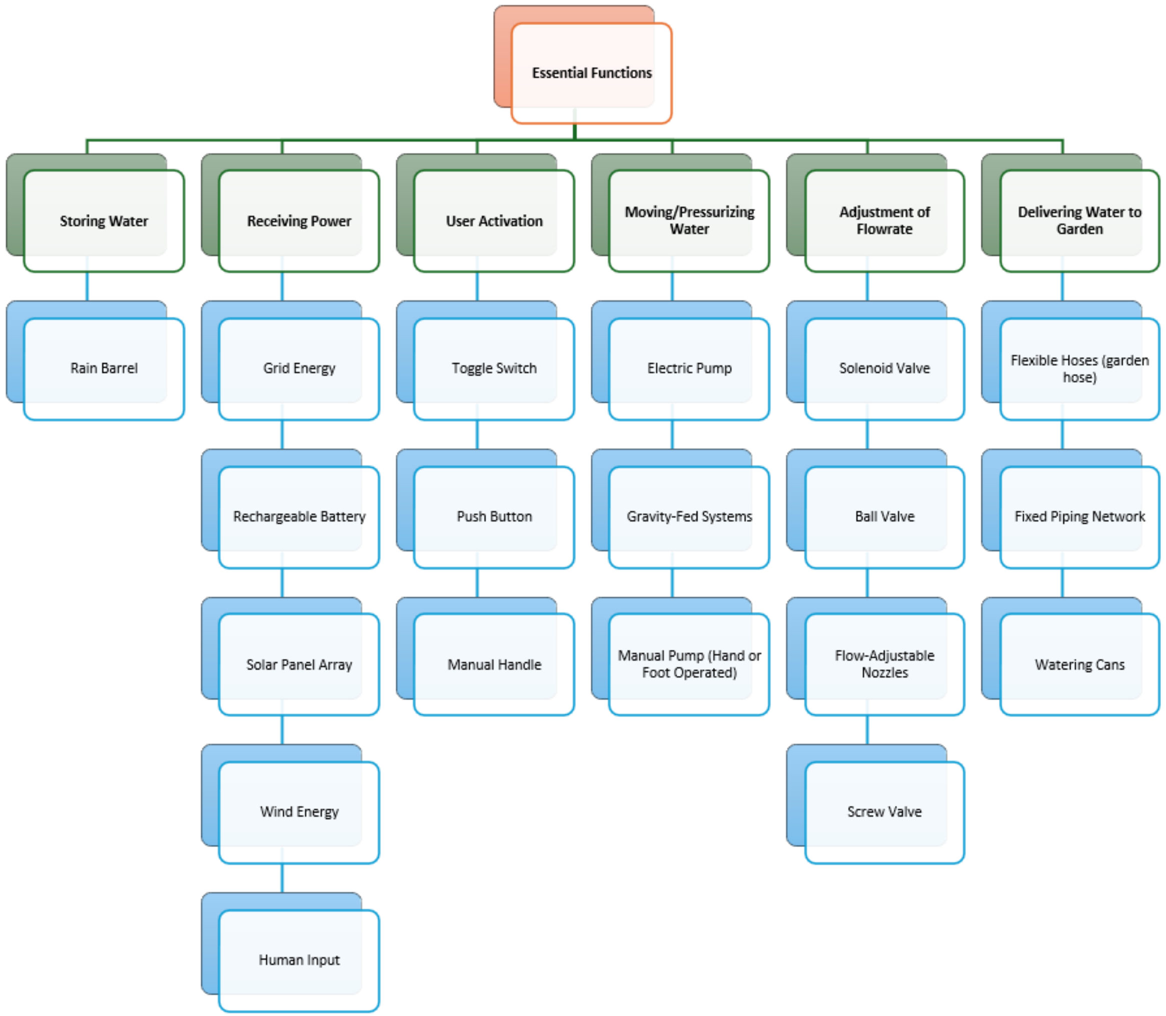
Functional Decomposition



Concept Combination

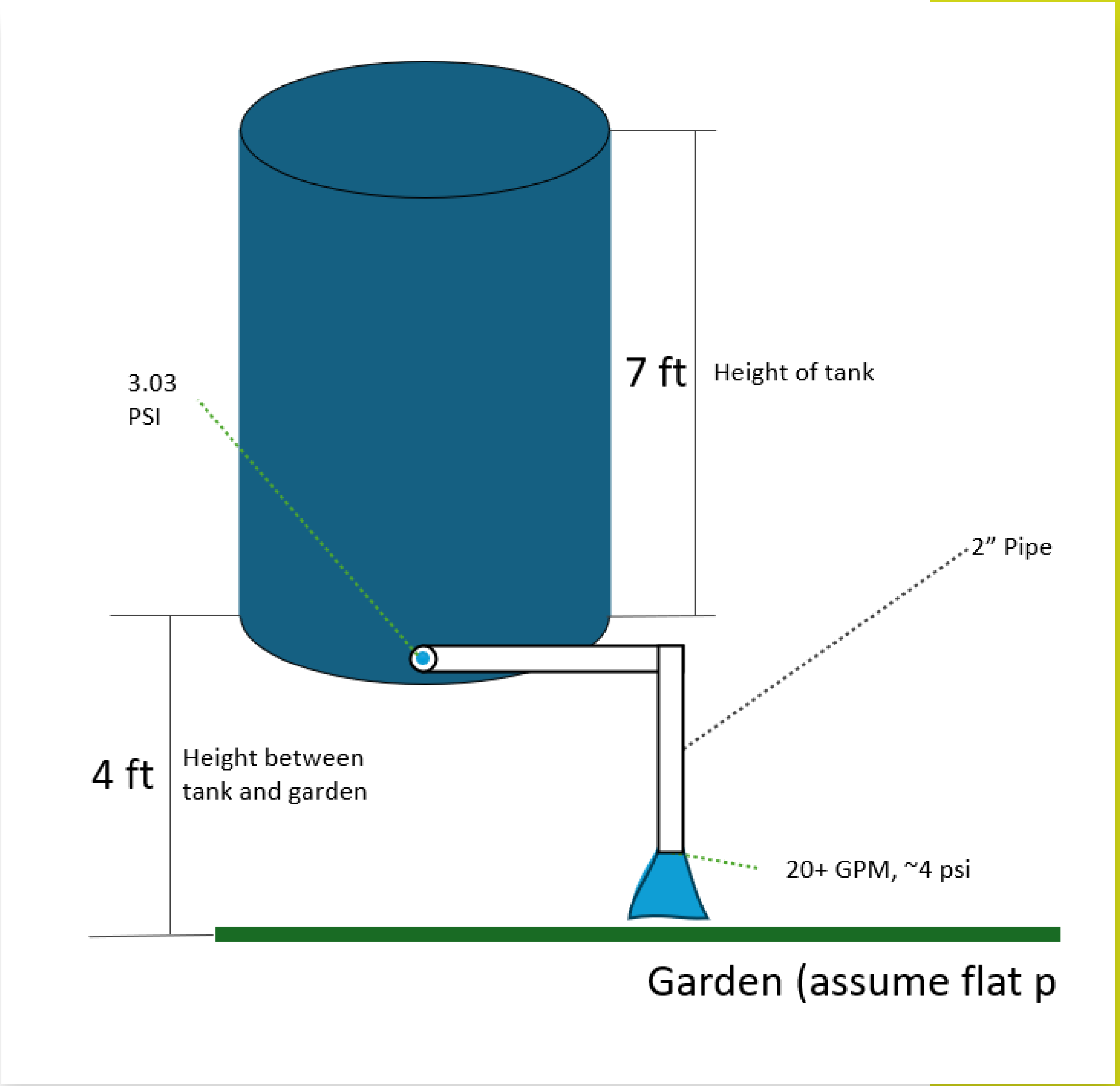
- Concept Combination is the process of mixing and matching different solutions for individual system functions to create an optimal overall design.
- The Concept Combination Table (CCT) serves as a systematic tool to explore these combinations, ensuring that every function is addressed and all innovative possibilities are considered

Concept Combination Table



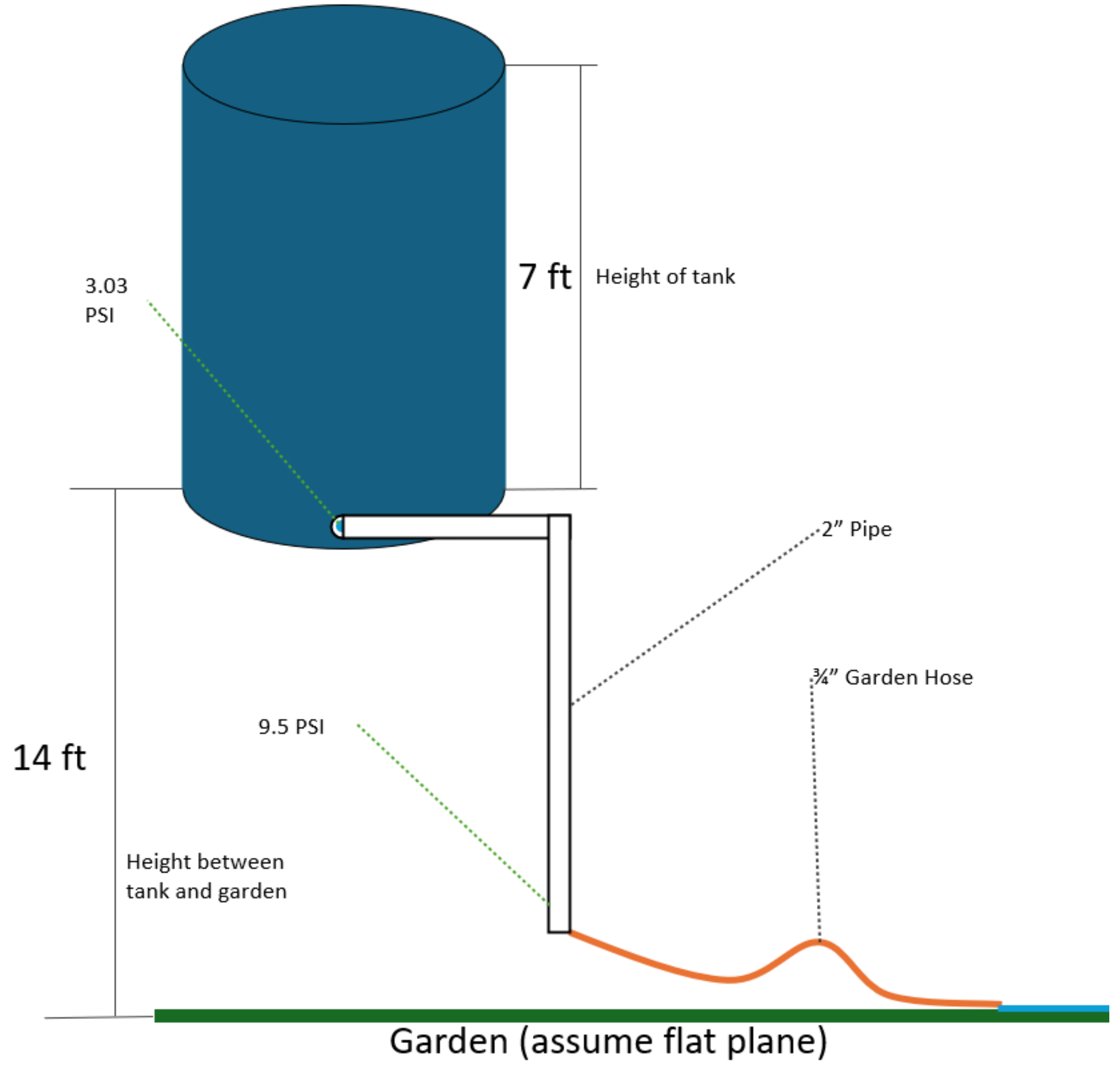
C1: Changing Pipe Diameter

- Good: Cheap. Easy to operate. Reliable
- Bad: No hose. Not versatile. Will not work with sprinkler. Limited capabilities



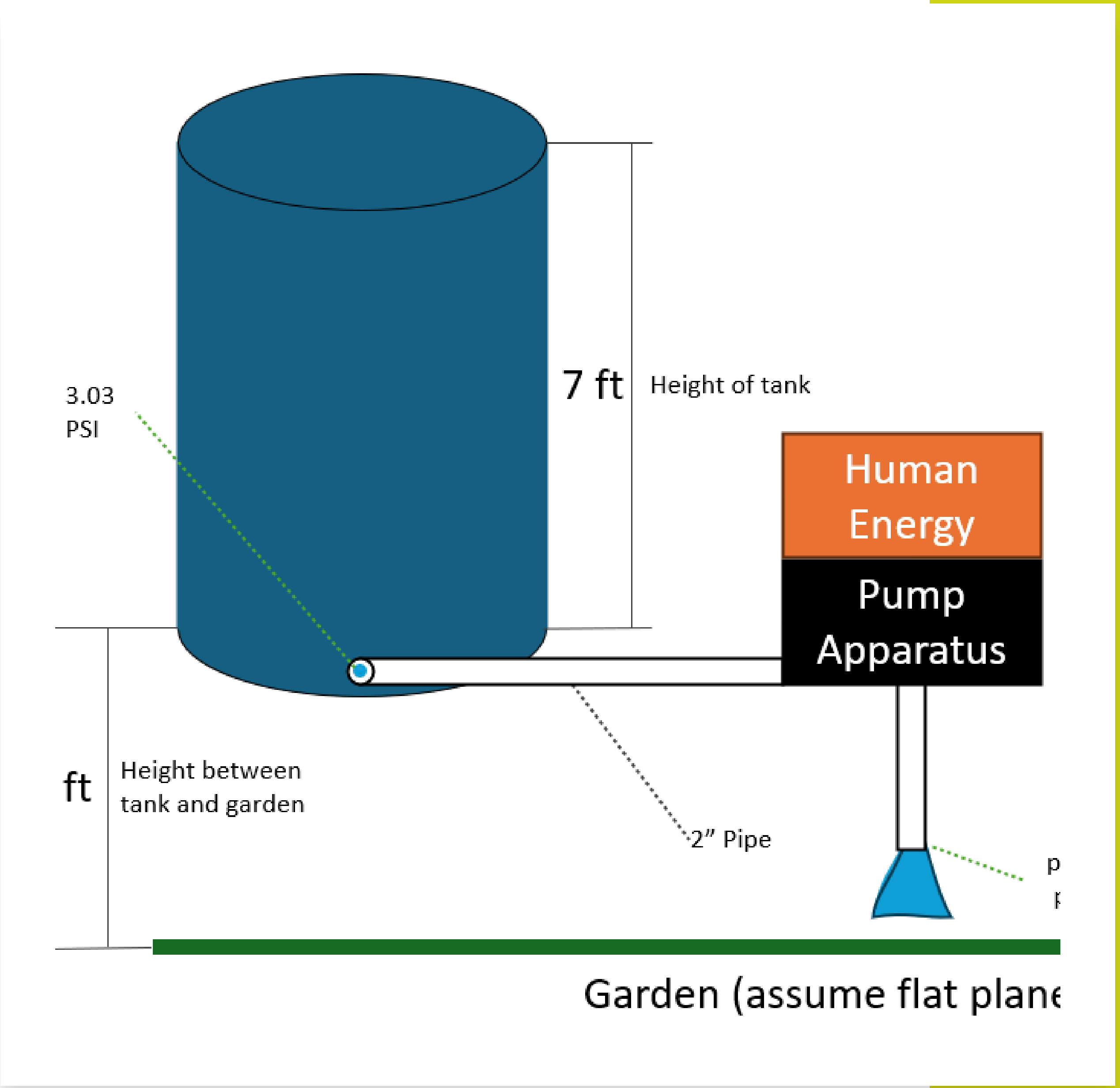
C2: Raising Tank

- Good: Cheap. Easy to operate. Reliable. Works with hose
- Bad: Raised Tank is ugly. Won't provide enough pressure to run sprinkler. Will be hard to do maintenance on



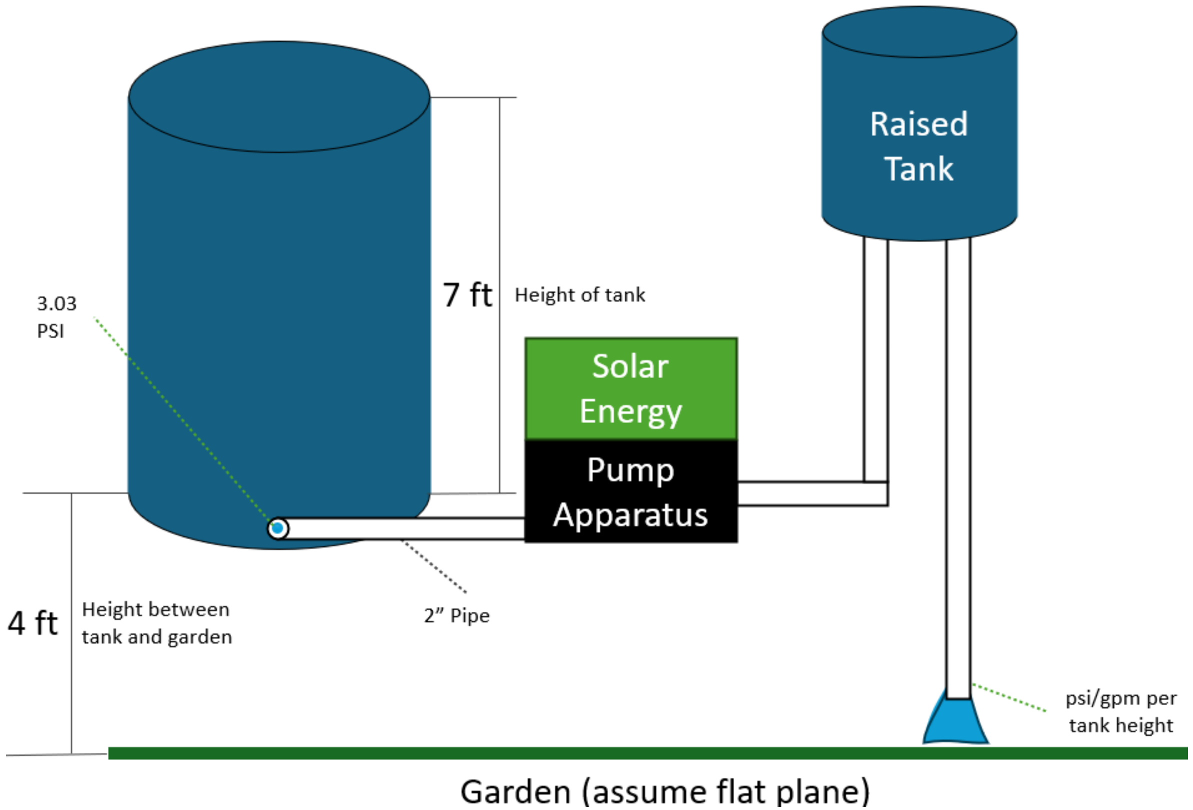
C3: Manual Pump

- Good: Simple to operate. Reliable. Easy to maintain
- Bad: Either low pressure or low flowrate. Requires manual labor. Medium maintenance



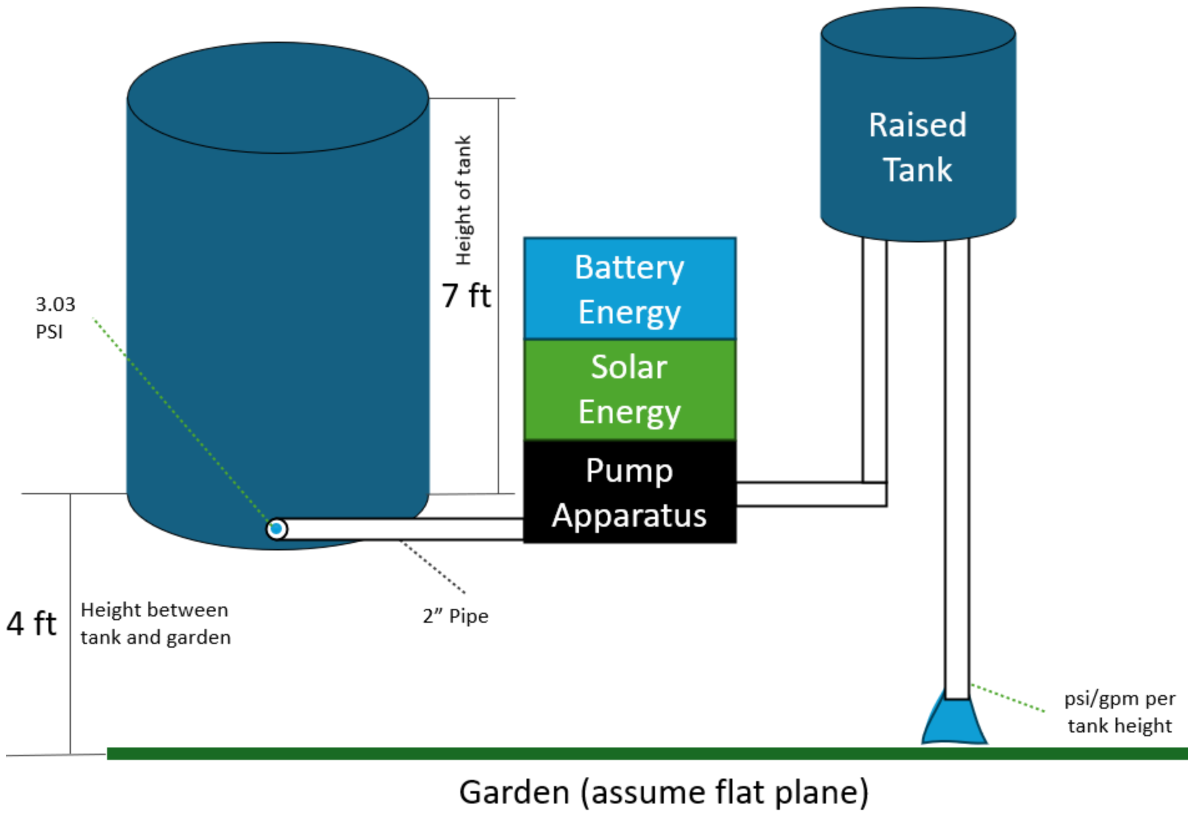
C4: Solar Pump + Raised Tank

- Good: No manual Labor. Good pressure.
- Bad: Raised tank is unsightly. Solar energy dependent on sun. More expensive.



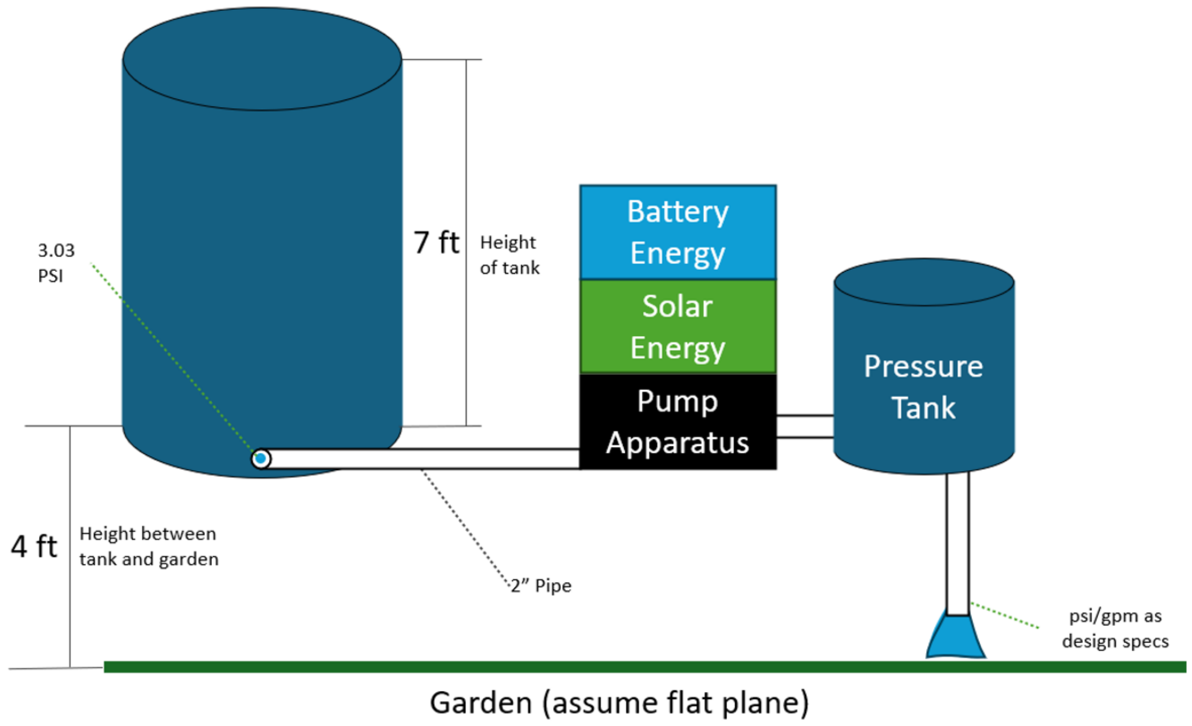
C5: Solar + Battery + Raised

- Good: No manual Labor. Good pressure. More reliable
- Bad: Raised tank is unsightly. Solar energy dependent on sun. More expensive.



C6: Solar + Battery + Pressure

- Good: No manual Labor. Best pressure. Mostly reliable
- Bad: Complex engineering. Most expensive. Hardest to maintain.



Summary of Viable Solutions

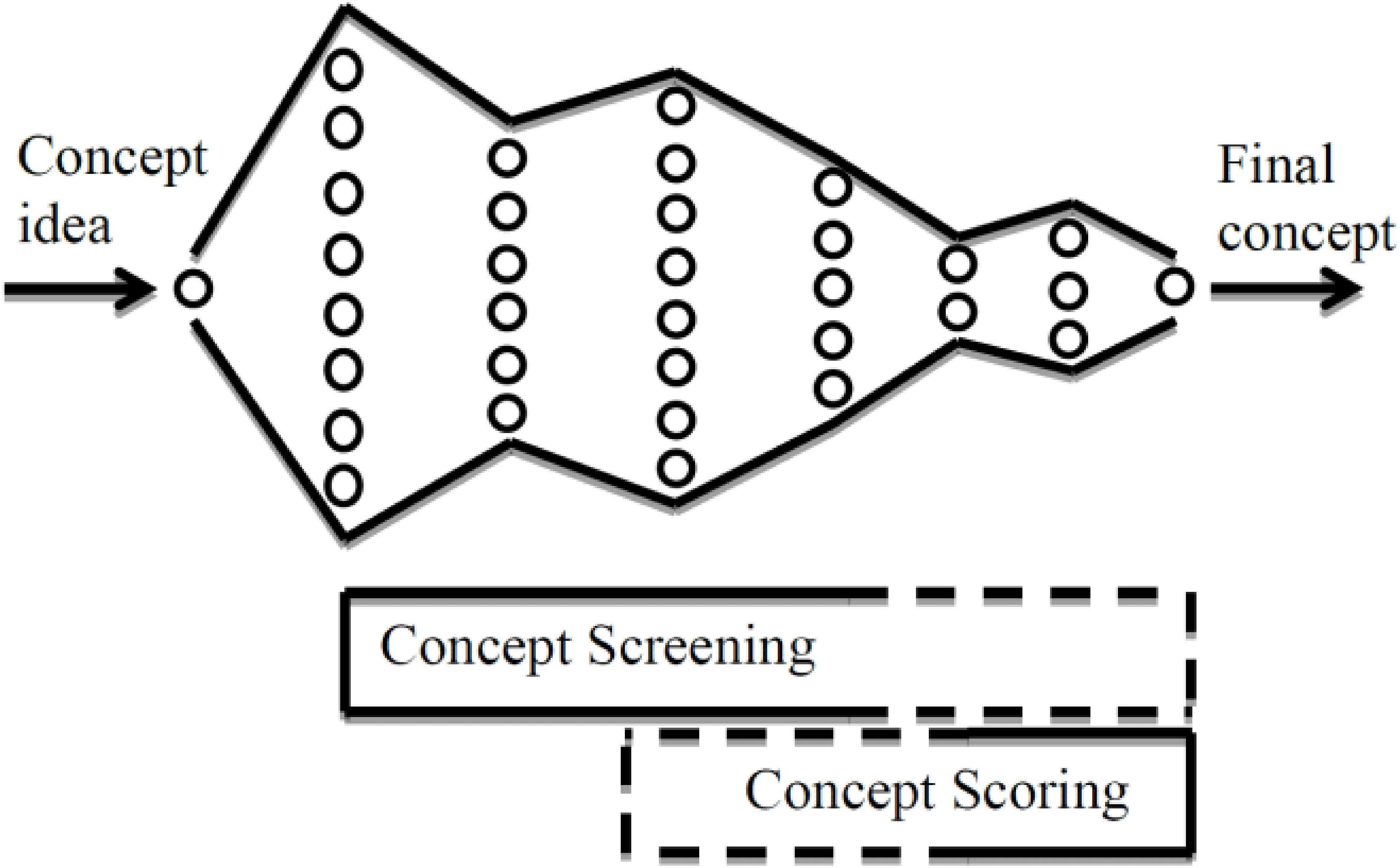
Concept Name	Advantages	Limitations
2-Inch Pipe System	- High flow rate (>20 GPM) - Rapid transfer - Low cost	- Incompatible with standard hoses
Gravity-Enhanced Tank	- Increased pressure (~10 PSI) - Simple operation	- Requires structural support
Manual Pump System	- Enhanced flow control - No external power needed	- Requires manual labor
Solar-Powered + Raised Tank	- Renewable energy use - Steady pressure	- Needs stable structure - Daylight dependent
Solar + Battery + Pressure Tank	- Consistent operation - Steady water flow	- High cost and complexity
Solar + Battery-Powered + Raised Tank	- Reliable delivery - Improved pressure	- Higher installation and maintenance costs - Aesthetic challenges

Concept Selection [H4]

Table 5: Raw Ratings of Selected Concepts

Criteria	**C1**	**C6**	**C9**	**C10**	**C12**	**C13**
Perceived Sustainability	3	5	4	4	5	5
Performance	3	3	4	5	5	5
Cost	5	3	3	2	1	2
Complexity	5	4	4	3	2	2
Safety	5	5	5	5	5	5
Reliability	5	4	4	4	4	4
User Experience	4	4	5	5	5	5

Concept Selection

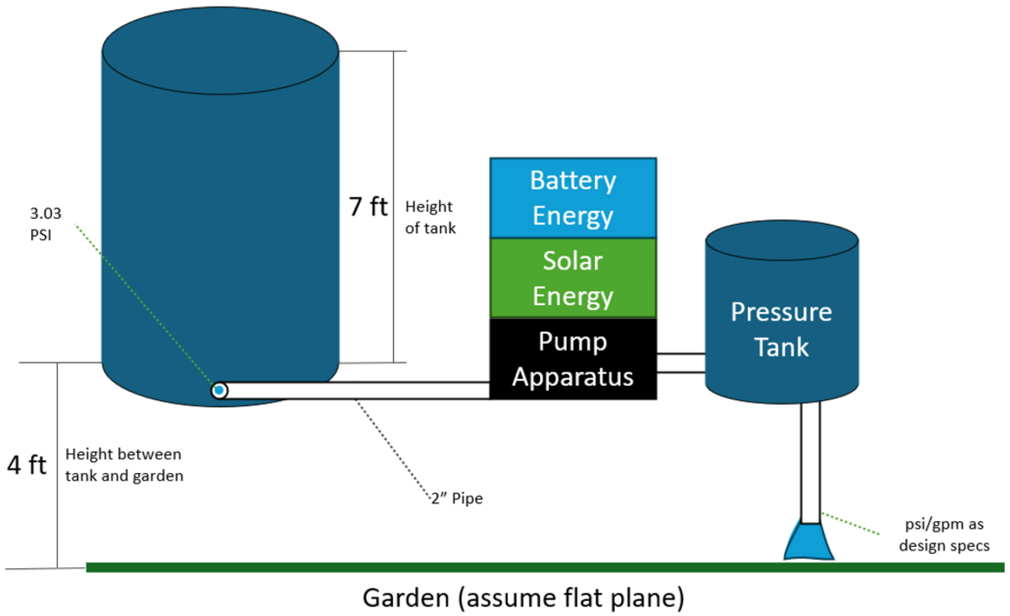


Concept Selection

Table 6: Weighted Scores of Selected Concepts

Criteria	**Weight**	**C1**	**C6**	**C9**	**C10**	**C12**	**C13**
Perceived Sustainability (36.7%)	0.367	1.10	1.83	1.47	1.47	1.83	1.83
Performance (20.5%)	0.205	0.62	0.62	0.82	1.03	1.03	1.03
Cost (11.1%)	0.111	0.55	0.33	0.33	0.22	0.11	0.22
Complexity (7.2%)	0.072	0.36	0.29	0.29	0.22	0.14	0.14
Safety (5.2%)	0.052	0.26	0.26	0.26	0.26	0.26	0.26
Reliability (4.3%)	0.043	0.22	0.13	0.17	0.17	0.17	0.17
User Experience (2.9%)	0.029	0.12	0.12	0.15	0.15	0.15	0.15
Total Score		**3.22**	**3.58**	**3.49**	**3.52**	**3.68**	**3.80**
Rank		6	4	5	3	2	1
Continue?							Yes

Top Concept



Sustainable and Reliable: Combines solar energy and battery storage for eco-friendly, uninterrupted operation.

Efficient Water Delivery: Pressurized tank ensures steady flow with minimal energy waste.

Cost-Effective: Long-term savings from renewable energy offset initial investment.

Adaptable and Scalable: Modular design supports future expansion and upgrades.

Refinements to the Top Concept

- **Cost Optimization**: Reuse of solar panels, batteries, and pump significantly reduced material and budget requirements.
- **Standardized Components**: Simplified maintenance with a uniform 2-inch PVC pipe design for easy repair and replacement.
- **Compact and Efficient Design**: Optimized layout of reused components to conserve space and facilitate straightforward installation.

End of Slides

References:

- Needs Report
- Product Specifications Report
- Functional Decomposition Report
- Concept Selection Report
- USC Office of Sustainability

Thanks for Listening!

William Wenzel
Presston McConnell
Tanner Harmon
Jacob Vaught